Classwork 1

1:CPU Functional Units

Objective

Understand the role of

- ALU
- Registers
- Control Unit
- Program Counter
- Instruction Register

Task

Write a simple ARM assembly program

to A = 25

B = 15

C = A + B

Must Observe

- Observe register changes
- Observe PC movement
- Observe instruction execution one by one

Deliverables

- Screenshot of Registers
- Explain ALU operation
- Explain PC after each instruction

2: CPU Fetch-Decode-Execute

Cycle Objective

Understand instruction

execution. **Program**

```
MOV R0,#10
MOV R1,#20
ADD R2,R0,R1
SUB R3,R2,#5
```

Fill the table

Step	PC	IR	Operation
1			
2			
3			
4			

3: Memory Organization

Objective

Understand memory

access. **Program**

```
LDR R0,=NUM
LDR R1,[R0]
ADD R1,R1,#10
STR R1,[R0]
NUM:
.word 100
```

Record

- Memory before execution
- Memory after execution •

Address accessed

- Data stored

4: Register Transfer

Operations Objective

Observe register

transfer**. Program**

```
MOV R0,#50
MOV R1,R0
MOV R2,R1
```

5: Bus Structure

Observation Objective

Understand

- Address Bus
- Data Bus
- Control Bus

Program

```
LDR R0,=DATA
LDR R1,[R0]
```